

Cart System API

Overview

This system allows customers to add products to cart, manage quantities, apply coupons, and move items to wishlist. Each cart belongs to a single customer and stores multiple cart items.

1. Cart Item Operations

Add item to cart

POST /cart/{cartId}/items

Purpose

Adds a product into the cart. If the product already exists, quantity can be increased.

Request body

```
{
  "product_id": 1,
  "quantity": 2
}
```

How data is handled

- System checks if product already exists in cart
- If yes → quantity is updated
- If no → new cart item is created

Response

```
{
  "message": "Item added to cart",
  "cart_id": 5
}
```

Delete item from cart

DELETE /cart/{cartId}/items/{itemId}

Purpose

Removes an item completely from the cart.

Response

```
{  
  "message": "Item removed from cart"  
}
```

Update cart item

PUT /cart/{cartId}/items/{itemId}

Purpose

Updates product or replaces item details.

Request body

```
{  
  "product_id": 2  
}
```

Response

```
{  
  "message": "Cart item updated"  
}
```

Update quantity

PUT /cart/{cartId}/items/{itemId}/quantity

Purpose

Changes quantity of a product in cart.

Request body

```
{  
  "quantity": 3  
}
```

Response

```
{  
  "message": "Quantity updated"  
}
```

2. Wishlist Operations

Move item to wishlist

POST /cart/{cartId}/items/{itemId}/move-to-wishlist

Purpose

Moves product from cart to wishlist.

Internal process

- Remove from cart_item table
- Insert into wishlist table

Response

```
{  
  "message": "Item moved to wishlist"  
}
```

Remove from wishlist

DELETE /wishlist/{wishlistId}

Purpose

Deletes item from wishlist.

Response

```
{  
  "message": "Item removed from wishlist"  
}
```

3. Category Filtering

Filter cart by category

GET /cart/{cartId}/items?category_id=2

Purpose

Returns only cart items belonging to a specific category.

How data is fetched

- Join Cart_Item with Product
- Filter by category_id

Response

```
{  
  "cart_id": 5,  
  "items": [  
    {  
      "item_id": 1,  
      "product_name": "Shirt",  
      "category": "Clothing",  
      "price": 1200,  
      "quantity": 2,  
      "total": 2400  
    }  
  ]  
}
```

```
]
}
```

4. Coupon System

Apply coupon

POST /cart/{cartId}/apply-coupon

Purpose

Applies discount to entire cart.

Request body

```
{
  "coupon_code": "SAVE10"
}
```

How it works

- System validates coupon
- Calculates discount on cart total
- Stores coupon_code in cart table

Response

```
{
  "message": "Coupon applied",
  "discount": 500,
  "final_total": 4500
}
```

Remove coupon

DELETE /cart/{cartId}/remove-coupon

Purpose

Removes applied discount from cart.

Response

```
{  
  "message": "Coupon removed",  
  "final_total": 5000  
}
```